

Customer Support Ticket Classification: Prompt Specification Suite

1. Objective

Design and benchmark 3 prompt architectures for automated customer ticket triage across Technical, Billing, Account Management, and Feature Request categories.

2. Prompt Architecture Specifications

Prompt 1: Few-Shot Strict JSON Classifier

- **System Prompt:** You are a senior customer support triage assistant. Categorize tickets into exactly one category: [Technical, Billing, Account, Feature_Request]. Return valid JSON only.
- **Few-Shot Examples:** 4 curated examples covering edge cases (e.g., billing disputes caused by login failures).

Prompt 2: Zero-Shot Executive Summarizer

- **System Prompt:** Summarize the core issue, customer sentiment, and urgency in 2 sentences.
- **Constraints:** Maximum 40 words, output format **Issue: ... | Sentiment: ... | Urgency: ...**

Prompt 3: Chain-of-Thought Escalation Detector

- **System Prompt:** Evaluate whether this ticket requires tier-2 human supervisor escalation by analyzing customer churn risk, legal exposure, and severe technical outages.